

Sources of the Magnetic Field, solved

Step-by-step solutions to every practice problem in Chapter 19. Try each problem on your own before you read its solution.

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P19.1

Wire 1 produces circular field lines around itself. At the position of wire 2, this field $\vec{B}_1$ is perpendicular both to the line joining the wires and to the wires themselves. Wire 2 carries current in the same direction as wire 1, so the force on it, $I_2 \vec{L} \times \vec{B}_1$, points along the line joining the wires, **towards wire 1**. You can check this with the right-hand grip rule for $\vec{B}_1$ and then the right-hand rule for the cross product.

By Newton's third law, wire 1 is pulled towards wire 2 just as hard, so the wires attract. Reversing one current reverses the force, so antiparallel currents repel.

P19.2

A solenoid is a stack of current loops. **Inside**, every loop's field points the same way along the axis, so the contributions add up. Far from the ends, the result is uniform, as the Ampère's-law calculation shows.

Outside, consider a point just beyond the windings. The nearby wires on that side of the solenoid carry current in one direction, while the wires on the far side carry it in the opposite direction. Their contributions largely cancel, and for a long, tightly wound solenoid the field outside approaches zero. The field lines return through the vast space outside, so they are spread very thin there.

P19.3

Ampère's law needs two things. First, a symmetric Amperian loop on which $\mathbf{B}$ is constant, so that it can be taken out of the integral. Second, a steady current that forms a complete circuit. A short segment provides neither. Its field is not uniform around any simple loop, and the segment is not a closed circuit: in reality, the rest of the circuit also contributes to the field. To find the field of the segment alone, we must use Biot–Savart (Problem P19.13).

P19.4

No. Above the Curie temperature, thermal agitation destroys the exchange alignment, and the material becomes paramagnetic. When it cools again with no external field, domains re-form, but their magnetisation directions are random. The net magnetisation is essentially zero. (A weak field present during cooling, such as the Earth's, leaves a slight net magnetisation. This is how volcanic rocks record the Earth's field at the time they cooled, the basis of *palaeomagnetism*.) To make a magnet again, the material must be re-magnetised in a strong field.

P19.5

Field at the midpoint, 0.050 m from each wire. With $\mu_0/2\pi = 2.0 \times 10^{-7}$ T m/A:

- From the 5.0 A wire: $B_1 = (2.0 \times 10^{-7})(5.0)/0.050 = 2.0 \times 10^{-5}$ T.
- From the 8.0 A wire: $B_2 = (2.0 \times 10^{-7})(8.0)/0.050 = 3.2 \times 10^{-5}$ T.

The currents are in the same direction, so between the wires the two fields point in **opposite** directions:

$$B = 3.2 \times 10^{-5} - 2.0 \times 10^{-5} = 1.2 \times 10^{-5} \text{ T},$$

in the direction of the larger wire's field.

Force per metre:

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} = (2.0 \times 10^{-7}) \frac{(5.0)(8.0)}{0.10} = 8.0 \times 10^{-5} \text{ N/m},$$

attractive.

P19.6

The fields cancel where $\frac{I_1}{x} = \frac{I_2}{d-x}$, with x measured from the 5.0 A wire:

$$\frac{5.0}{x} = \frac{8.0}{0.10 - x} \Rightarrow 0.50 - 5.0x = 8.0x \Rightarrow x = 0.038 \text{ m.}$$

The field is zero 3.8 cm from the 5.0 A wire, closer to the weaker current, as expected.

P19.7

$$B = \frac{\mu_0 I}{2\pi r} = \frac{(2.0 \times 10^{-7})(500)}{15} = 6.7 \times 10^{-6} \text{ T} = 6.7 \mu\text{T}.$$

This is about 13% of the Earth's static field, and only about 3% of the 200 μT guideline. Real lines carry three-phase currents in several conductors whose fields partly cancel, so the measured fields are usually even smaller. Large epidemiological studies have not established any health effects at such exposures.

P19.8

At the centre:

$$B_0 = \frac{\mu_0 NI}{2R} = \frac{(4\pi \times 10^{-7})(100)(1.0)}{2(0.10)} = 6.3 \times 10^{-4} \text{ T}.$$

On the axis at $x = R$:

$$B = B_0 \frac{R^3}{(2R^2)^{3/2}} = \frac{B_0}{2\sqrt{2}} = 2.2 \times 10^{-4} \text{ T}.$$

P19.9

Field: $n = 1000/0.50 = 2000$ turns/m, so

$$B = \mu_0 n I = (4\pi \times 10^{-7})(2000)(2.0) = 5.0 \times 10^{-3} \text{ T}.$$

Flux:

$$\Phi = BA = (5.03 \times 10^{-3})(4.0 \times 10^{-4}) = 2.0 \times 10^{-6} \text{ Wb}.$$

P19.10

- At 0.50 mm (inside the inner conductor). Only a fraction $(r/R)^2$ of the current is enclosed:

$$B = \frac{\mu_0 I r}{2\pi R^2} = (2.0 \times 10^{-7}) \frac{(5.0)(5.0 \times 10^{-4})}{(1.0 \times 10^{-3})^2} = 5.0 \times 10^{-4} \text{ T.}$$

- At 2.0 mm (between the conductors). The full 5.0 A is enclosed:

$$B = (2.0 \times 10^{-7}) \frac{5.0}{2.0 \times 10^{-3}} = 5.0 \times 10^{-4} \text{ T.}$$

(The two values happen to be equal because $0.5 \times 2.0 = 1.0^2$, the same coincidence as in Problem P14.8.)

- At 10 mm (outside the cable). The net enclosed current is zero, so $B = 0$.

P19.11

Field at the electron:

$$B = \frac{\mu_0 I}{2\pi r} = (2.0 \times 10^{-7}) \frac{20}{0.020} = 2.0 \times 10^{-4} \text{ T.}$$

Force:

$$F = evB = (1.602 \times 10^{-19})(1.0 \times 10^6)(2.0 \times 10^{-4}) = 3.2 \times 10^{-17} \text{ N.}$$

Direction. An electron moving along the direction of the current is equivalent to a current flowing the *opposite* way. Antiparallel currents repel, so the force pushes the electron **away from the wire**.

P19.12

- (a) **Rate of change of the field.** The peak field is 1.35 T (Example 19.6):

$$\frac{dB}{dt} \approx \frac{1.35 \text{ T}}{1.0 \times 10^{-4} \text{ s}} = 1.3 \times 10^4 \text{ T/s.}$$

This enormous rate of change is what induces the stimulating currents in the brain.

- (b) **Force between adjacent turns, which carry parallel currents:**

$$\frac{F}{L} = \frac{\mu_0 I^2}{2\pi d} = (2.0 \times 10^{-7}) \frac{(5.0 \times 10^3)^2}{0.0040} = 1.3 \times 10^3 \text{ N/m.}$$

That is about 130 kg-weight per metre, applied in a sudden jerk thousands of times per session. TMS coils must be mechanically very strong and potted in resin. The pulses also make a loud click, so patients wear ear protection. In addition, the coils heat up and need cooling during long sessions.

P19.13

Derivation. Put the wire along the x -axis and the field point at perpendicular distance d from it, with the foot of the perpendicular at $x = 0$. An element dx at position x is a distance $r = \sqrt{d^2 + x^2}$ from the field point. The angle between $d\vec{l}$ and $\hat{r}$ has sine d/r , so

$$dB = \frac{\mu_0 I}{4\pi} \frac{dx (d/r)}{r^2} = \frac{\mu_0 I}{4\pi} \frac{d dx}{(d^2 + x^2)^{3/2}}.$$

Substitute $x = d \tan \alpha$, so that $dx = d \sec^2 \alpha d\alpha$ and $(d^2 + x^2)^{3/2} = d^3 \sec^3 \alpha$:

$$dB = \frac{\mu_0 I}{4\pi d} \cos \alpha d\alpha \Rightarrow B = \frac{\mu_0 I}{4\pi d} (\sin \alpha_2 - \sin \alpha_1).$$

(a) **Infinite wire:** $\alpha_1 = -90^\circ$ and $\alpha_2 = +90^\circ$, so $B = \frac{\mu_0 I}{4\pi d} (2) = \frac{\mu_0 I}{2\pi d}$.

(b) **Square loop.** Each side is at $d = a/2$ from the centre, with α running from -45° to $+45^\circ$:

$$B_{\text{side}} = \frac{\mu_0 I}{4\pi(a/2)} (2 \sin 45^\circ) = \frac{\sqrt{2} \mu_0 I}{2\pi a}.$$

All four sides contribute in the same direction:

$$B = \frac{2\sqrt{2} \mu_0 I}{\pi a} = 0.90 \frac{\mu_0 I}{a}.$$

Circle with the same perimeter ($2\pi R = 4a$, so $R = 2a/\pi$):

$$B = \frac{\mu_0 I}{2R} = \frac{\pi \mu_0 I}{4a} = 0.79 \frac{\mu_0 I}{a}.$$

The square gives about 15% more field at its centre, because its sides come closer to the centre (a distance $a/2 = 0.5a$, against $R = 0.64a$).

P19.14

Semicircle: this is an arc with $\phi = \pi$, so it contributes $B = \mu_0 I \pi / (4\pi R) = \mu_0 I / (4R)$.

Straight sections. Each long straight section starts level with the centre and extends to infinity, at a perpendicular distance R . Using P19.13 with α running from 0 to 90° , each gives

$$B = \frac{\mu_0 I}{4\pi R}.$$

The two straight sections carry current in opposite directions on opposite sides of the centre, so their fields point the same way. All three contributions are in the same direction, perpendicular to the plane of the hairpin:

$$B = \frac{\mu_0 I}{4R} + 2 \cdot \frac{\mu_0 I}{4\pi R} = \frac{\mu_0 I}{4R} \left(1 + \frac{2}{\pi} \right).$$

P19.15

(a) **Field at the midpoint.** Each coil is $R/2$ from the midpoint, and both contribute in the same direction:

$$B = 2 \cdot \frac{\mu_0 N I R^2}{2(R^2 + R^2/4)^{3/2}} = \frac{\mu_0 N I R^2}{(5/4)^{3/2} R^3} = \left(\frac{4}{5} \right)^{3/2} \frac{\mu_0 N I}{R} = 0.716 \frac{\mu_0 N I}{R}.$$

(b) **Uniformity.**

- *First derivative:* by symmetry, $dB/dx = 0$ at the midpoint. The rise in one coil's field as you move towards it is exactly matched by the fall in the other's.
- *Second derivative:* a single coil's field is proportional to $f(x) = (x^2 + R^2)^{-3/2}$. Differentiating twice gives $f''(x) = 3(4x^2 - R^2)(x^2 + R^2)^{-7/2}$, which is zero at $x = R/2$. Each coil is exactly $R/2$ from the midpoint, so both second derivatives vanish there.

The first non-zero correction is fourth order, so the field is uniform to within about 1% over a large central region. This is why the spacing is chosen to equal the radius.

(c) **Numbers:**

$$B = 0.716 \times \frac{(4\pi \times 10^{-7})(100)(2.0)}{0.30} = 6.0 \times 10^{-4} \text{ T}.$$

Uses: cancelling the Earth's field around sensitive experiments; calibrating magnetometers; the classic e/m electron-beam experiment; exposing cell cultures to well-controlled magnetic fields in biological research.

P19.16

(a) Total current:

$$I = \int_0^R J_0 \frac{r}{R} 2\pi r dr = \frac{2\pi J_0}{R} \cdot \frac{R^3}{3} = \frac{2\pi J_0 R^2}{3}.$$

(b) Field.

- Inside: the enclosed current is $\frac{2\pi J_0 r^3}{3R} = I \frac{r^3}{R^3}$, so

$$B(2\pi r) = \mu_0 I \frac{r^3}{R^3} \Rightarrow B = \frac{\mu_0 I r^2}{2\pi R^3}.$$

- Outside: $B = \frac{\mu_0 I}{2\pi r}$.

The two agree at $r = R$. ✓ Inside, the field now grows as r^2 rather than r , because the current is concentrated towards the surface. (At high frequencies, current crowds into the outer layer of a conductor, the *skin effect*, giving exactly this kind of distribution.)

P19.17

(a) **Derivation.** Use the field $H = B/(\mu_r \mu_0)$, for which Ampère's law reads $\oint H dl = NI$. If B is the same in the iron and in the gap (the flux is continuous):

$$\frac{B}{\mu_r \mu_0} L + \frac{B}{\mu_0} g = NI \Rightarrow B = \frac{\mu_0 NI}{g + L/\mu_r}.$$

(b) Field in the gap. $L/\mu_r = 0.30/2000 = 1.5 \times 10^{-4}$ m:

$$B = \frac{(4\pi \times 10^{-7})(500)(2.0)}{2.0 \times 10^{-3} + 1.5 \times 10^{-4}} = \frac{1.26 \times 10^{-3}}{2.15 \times 10^{-3}} = 0.58 \text{ T}.$$

(c) No gap:

$$B = \frac{\mu_r \mu_0 NI}{L} = \frac{2000(4\pi \times 10^{-7})(1000)}{0.30} = 8.4 \text{ T}.$$

Iron would actually saturate at about 2 T, so the real field would be limited to that.

Comment. A 2 mm gap in a 300 mm iron path dominates the whole “magnetic circuit”. It acts like a large resistance in series with a small one, since the reluctance $\ell/(\mu_r \mu_0 A)$ of air is μ_r times that of iron. Gaps are deliberately built into inductors and motors: they make the field depend

predictably on the current, prevent saturation, and the gap is where the useful energy is stored. In loudspeakers, motors and MRI-compatible devices, the gap is where the work gets done.

P19.18

(a) **Equivalent current.** The electron passes any point on its orbit $v/(2\pi r)$ times per second:

$$I = \frac{ev}{2\pi r} = \frac{(1.602 \times 10^{-19})(2.19 \times 10^6)}{2\pi(5.29 \times 10^{-11})} = 1.06 \times 10^{-3} \text{ A.}$$

(b) **Magnetic moment:**

$$\mu = I\pi r^2 = (1.06 \times 10^{-3})\pi(5.29 \times 10^{-11})^2 = 9.27 \times 10^{-24} \text{ A m}^2.$$

In general, $\mu = \frac{ev}{2\pi r}\pi r^2 = \frac{evr}{2} = \frac{e}{2m_e}(m_e vr)$. In the Bohr model the angular momentum is $m_e vr = \hbar$, so $\mu = e\hbar/2m_e = \mu_B$. ✓

(c) **Field at the proton** (the centre of the loop):

$$B = \frac{\mu_0 I}{2r} = \frac{(4\pi \times 10^{-7})(1.06 \times 10^{-3})}{2(5.29 \times 10^{-11})} = 12.5 \text{ T.}$$

That is stronger than a research MRI magnet, produced by a single orbiting electron. Such internal atomic fields split spectral lines (fine structure), and the interaction between the electron's field and the proton's own magnetic moment gives hydrogen's famous 21 cm radio line, which radio astronomers use to map the galaxy.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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